MIDTERM EXAMINATION
INTEGRATIVE PROGRAMMING AND TECHNOLOGIES
First Semester, A.Y. 2026-2027

NAME:_____ YEAR&SECTION:_____ DATE:_____ SCORE:__/50

I. **MULTIPLE CHOICE:** Read each question carefully. Choose the letter of the correct answer from the given choices**.** Encircle the letter of your chosen answer**.** Make sure that your answer is clear and legible. Answer all items.

1. Which statement correctly classifies the relationship between the Internet and the Web?
   A. Web is physical infrastructure, Internet is software
   B. Internet is for browsers, Web is hardware
   C. They are the same layer
   D. Internet is physical infrastructure, Web is a software service
2. Why do domain names exist if computers route traffic using IP addresses?
   A. They replace IPs entirely
   B. They provide readable names for humans, while DNS translates them to IPs
   C. They exist purely for legal trademark protection
   D. IPs only work for email servers
3. What is the correct functional sequence when a browser loads a webpage?
   A. Server response, DNS query, GET request, packet routing
   B. GET request, packet routing, DNS query, server response
   C. DNS query, GET request, server response, packet routing
   D. Packet routing, GET request, DNS query, server response
4. What are the roles of the Header and Payload in a data packet?
   A. They contain identical data for redundancy
   B. Payload contains routing, Header contains data
   C. Header is encrypted content, Payload is the IP
   D. Header contains routing info, Payload contains actual data

5. Which part of a URL should be edited to change the requested resource without altering the targeted server?
   A. Protocol
   B. Path
   C. Subdomain
   D. Domain
6. What happens if a site uses relative URLs for images (e.g., "/images/logo.png") and migrates to a new domain?
   A. They will continue to work without modification
   B. They work only if the IPs match
   C. They will break immediately
   D. They will break due to a missing protocol
7. Which URL addressing strategy for images prevents unauthorized reuse and search engine duplication?
   A. Relative URLs
   B. Absolute URLs
   C. Relative URLs (cannot be externally linked)
   D. Absolute URLs (speeds up dev)
8. What is the most likely rendering outcome if a page omits the <!DOCTYPE html> declaration?
   A. Browser falls back to Quirks Mode
   B. Browser refuses to render
   C. Browser auto-inserts a default DOCTYPE
   D. Browser ignores the missing DOCTYPE
9. What is the effect of moving a <meta> tag before the DOCTYPE declaration?



A. Applies globally to the domain
B. No effect
C. Risks triggering Quirks Mode
D. Improves SEO

10. Why does visible text placed inside the <head> section fail to appear at the top of the page as expected?
A. <head> is invisible and for metadata; visible content belongs in <body>
B. It requires CSS classes to be visible
C. It only renders on mobile
D. It renders after a delay

11. What is the consequence of missing a <title> tag on a webpage?
A. Browser tab and search results lack a title
B. Page fails to load
C. Stylesheet stops applying
D. Character encoding defaults to an unsupported format

12. What is the likely cause of a webpage rendering as a tiny, unscaled desktop layout on a mobile phone?
A. Missing `<title>` tag
B. Missing `<meta name="viewport">` tag
C. Missing closing `</body>` tag
D. Extra charset tag

13. What is the consequence of placing a large <style> block after the closing </body> tag?
A. Stylesheet applies before HTML loads
B. Stylesheet moves into <head> automatically
C. Page fails to render
D. Styles may apply late, causing a flash of unstyled content

14. Based on separating system content from user content, where do <link rel="stylesheet"> and <h1> belong?
A. Link in <body>, <h1> in <head>
B. Link in <head>, <h1> in <body>
C. Both in <head>
D. Both in <body>

15. Why is skipping a heading rank (e.g., <h1> directly to <h3>) an accessibility concern?
A. Cosmetic issue only
B. Causes search engine penalties
C. Breaks the logical outline used by screen readers
D. Browser hides the content

16. How should heading ranks be structured within a fixed sidebar?
A. Consistent across all pages
B. One level lower than main content
C. Avoid headings entirely
D. Always start at <h1>

17. What is the practical difference between <b> and tags?
A. `<strong>` communicates semantic importance, `<b>` is only visual
B. `<b>` is semantic, `<strong>` is visual
C. Both are ignored by search engines
D. No practical difference

18. How should a developer bold a product name while marking it semantically important?
A. Use <i> alone
B. Use `<strong>` and apply CSS for extra visual styling
C. Use `<b>` alone
D. Use red font color only



19. What is the security vulnerability of using target="_blank" without additional attributes?
    A. Browsers block external links
    B. Prevents HTTPS loading
    C. The new tab can hijack the original page; fix with rel="noopener"
    D. Only affects internal links
20. Which VS Code feature instantly generates HTML boilerplate from a single character?
    A. Debugger
    B. Prettier
    C. Live Server
    D. Emmet
21. Which VS Code extension automatically refreshes the browser when code is saved?
    A. Prettier
    B. Live Server
    C. Emmet
    D. Debugger
22. Which VS Code feature is best for pausing a script to inspect a live variable value?
    A. Command Palette
    B. Prettier
    C. Minimap
    D. Debugger with a breakpoint
23. Which tool automatically corrects code formatting and indentation upon saving?
    A. Prettier
    B. Emmet
    C. Live Server
    D. Debugger
24. Which stage of the web lifecycle is jeopardized by omitting the DOCTYPE?
    A. The Request stage
    B. The Maker stage
    C. The Server & Network stage
    D. The Render stage (risks Quirks Mode)
25. Why does mismatched capitalization like <Div>Content</DIV> still render correctly?
    A. HTML5 auto-converts to lowercase
    B. Works only for empty elements
    C. Browser silently ignores it
    D. HTML tags are not case sensitive
26. Why is
</br> an incorrect way to write a line break?
    A. It is actually correct and necessary
    B. `<br>` is an empty element and needs no closing tag
    C. It causes double line breaks
    D. It requires width and height attributes
27. Which example correctly follows the standard HTML element pattern (start tag, content, end tag)?
    A. <h1>My First Heading</h1>
    B.
    C. <!DOCTYPE html>
    D. <img src="photo.jpg">
28. What happens if an attribute is placed outside the start tag (e.g., <p>class="highlight">)?
    A. Applies to the next sibling
    B. Applies correctly
    C. Fails to render entirely
    D. Not applied; treated as visible text
29. Which attribute specifies a hyperlink's destination, and where must it be placed?



A. alt in the start tag
B. src in the start tag
C. href in the closing tag
D. href in the start tag

30. What happens if a broken image lacks an alt attribute?
A. Page refuses to render
B. Image is replaced by filename
C. Sighted users see a broken icon, screen reader users get no description
D. Screen readers read the file path

31. Which approach correctly applies red text to a single paragraph inline?
A. style attribute
B. alt attribute
C. title attribute
D. Wrapping in an <a> tag

32. Which HTML attribute creates a hover tooltip?
A. src
B. style
C. title
D. alt

33. How does a valid document containing <html><body> Hello </body></html> but missing a DOCTYPE align with best practices?
A. Fully correct
B. Incorrect because `<html>` must be inside `<body>`
C. Fails to render
D. Renders, but does not follow best practice as it risks Quirks Mode

34. Why do multiple spaces in an HTML source code paragraph render as a single space?
A. Error fallback behavior
B. Browsers collapse extra whitespace by default
C. DOCTYPE is missing
D. Converted to `<br>` tags

35. Which tag creates a horizontal line and requires no closing tag?
A. <hr></hr>
B.
C. <hr>
D.

36. Which tag forces a line break inside a paragraph without adding extra vertical spacing?
A. New `<p>` tag
B. `<hr>` tag
C. `<br>` tag
D. `<title>` tag

37. Which tag correctly adds a favicon to a webpage?
A. <img> in <body>
B. <link> in <head>
C. <meta> in <head>
D. icon attribute in <title>

38. Which tag combination creates an automatically numbered list?
A. and
B. <dl>, <dt>, and <dd>
C. and
D. and



39. How do you change an ordered list to use uppercase Roman numerals?
    A. Add type="I" to
    B. Add type="i" to
    C. Use <ul type="I">
    D. Add style="numbering:roman;" to <body>
40. Which structure is most appropriate for a glossary pairing terms with definitions?
    A. Unordered list (`<ul>`)
    B. Ordered list (`<ol>`)
    C. Description list (`<dl>`, `<dt>`, `<dd>`)
    D. Table (`<table>`)
41. Which tag specifically designates a table cell as a column header?
    A. <th>
    B. <td>
    C. <tr>
    D. <thead>
42. What is the assessment of nesting a <td> tag directly inside another <td> tag?
    A. Valid for merged cells
    B. Valid if headers are present
    C. Structurally invalid
    D. Valid due to free nesting rules
43. Which inline tag correctly formats a mathematical exponent (e.g., mc²)?
    A. <small>
    B. <mark>t;
    C. <sub>
    D. <sup>
44. Which tag appropriately reduces text size for a legal disclaimer?
    A. `<small>` (reduces size) is appropriate; `<mark>` highlights relevance
    B. `<mark>` is standard
    C. `<sup>` is standard
    D. Both are identical
45. Which tag lowers text for a chemical formula (e.g., H2O)?
    A. <small>
    B. `<sub>` (lowers text) vs `<sup>` (raises text)
    C. Both are identical
    D. <sup>
46. Which fix prioritizes displaying a readable label in the browser tab?
    A. Adding the `<title>` tag
    B. Adding viewport meta
    C. Removing visible text from `<head>`
    D. Moving CSS out of body
47. Which sequence represents a logical instructional progression for HTML concepts?
    A. Tables -> attributes -> document structure -> headings
    B. Links/images -> lists/tables -> document structure
    C. Document structure -> headings/paragraphs -> inline formatting -> lists/links
    D. Inline formatting -> document structure -> attributes -> headings
48. Which code combination fulfills all requirements for a profile page with a title, header, accessible image, and secure link?
    A. DOCTYPE, <title> in <body>, <img> with src, <a target="_blank">



B. DOCTYPE, `<title>` in `<head>`, `<h1>`, `<img>` with `alt`, `<a target="_blank" rel="noopener">`

C. <h1> in <head>, <img> with alt

D. DOCTYPE, <img> with style, <a> with alt

49. How should external links be formatted to open safely in a new tab and be indexed effectively?

A. rel="noopener" without target="_blank"

B. target="_blank" without rel="noopener"

C. Use `<img>` tags

D. Use `target="_blank"`, `rel="noopener"`, and descriptive text

50. Which navigation design survives domain migration, provides accessible titles, and shows a consistent favicon?

A. Absolute URLs, <title> in <head>, favicon in <body>

B. Tables cells, favicon in <body>

C. Relative URLs, <title> and <link rel="icon"> in <head>

D.Ordered list of <img>, <title> in <body>

"Integrity is doing the right thing, even when no one is watching." — C.S. Lewis

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Answer Key (1-50):

1. D
2. B
3. C
4. D
5. B
6. A
7. B
8. A
9. C
10. A
11. A
12. B
13. D
14. B
15. C
16. A
17. A
18. B
19. C
20. D
21. B
22. D
23. A
24. D
25. D
26. B
27. A
28. D
29. D
30. C
31. A
32. C
33. D
34. B
35. C
36. C
37. B
38. D
39. A
40. C
41. A
42. C
43. D
44. A
45. B
46. A
47. C
48. B
49. D
50. C

